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Lesson 7: Trees and Ensembles

Technologies for Artificial Intelligence

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Agenda

- A fourth family, built on a threshold
- Decision trees, and why one alone is unstable
- Depth as the bias-variance dial, again
- Bagging and random forests
- Feature importance, and its limits
- Gradient boosting
- Choosing among trees, forests and boosting

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Exercise 6, before we start

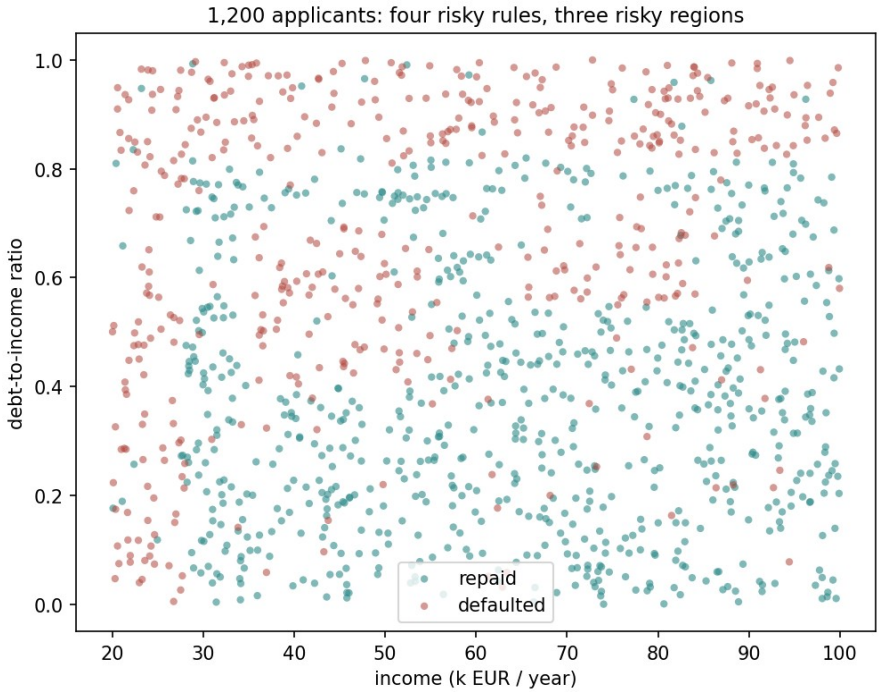
- What counts is the **defence of which family**, not the winning score
- The gap to watch for: a strong accuracy number with no check of *why* the assumption behind it held
- Today’s models come with the same obligation

1,200 loan applicants, two figures each

- Income in thousands of euros a year, and debt-to-income ratio
- Underwriting is a **checklist**: an income floor, a debt ceiling, and two “stressed” combinations, each figure ordinary alone but risky together
- **7%** of labels flipped: no honest model beats about **0.93**, the **noise ceiling**
- **38.7%** defaulted, so the majority baseline is **0.613**

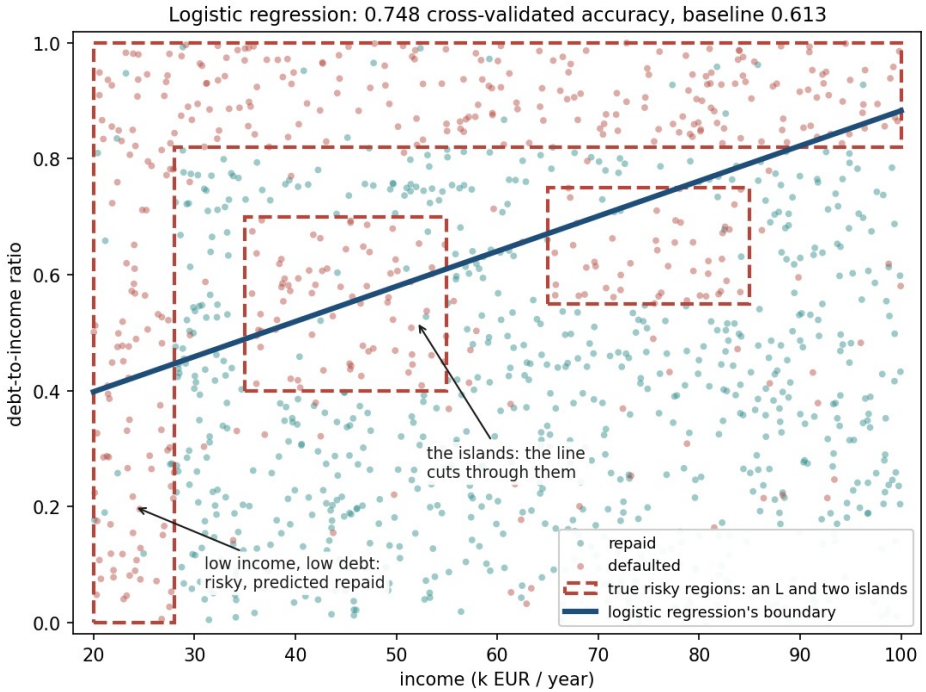
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Four risky rules, three risky regions



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No straight line separates them



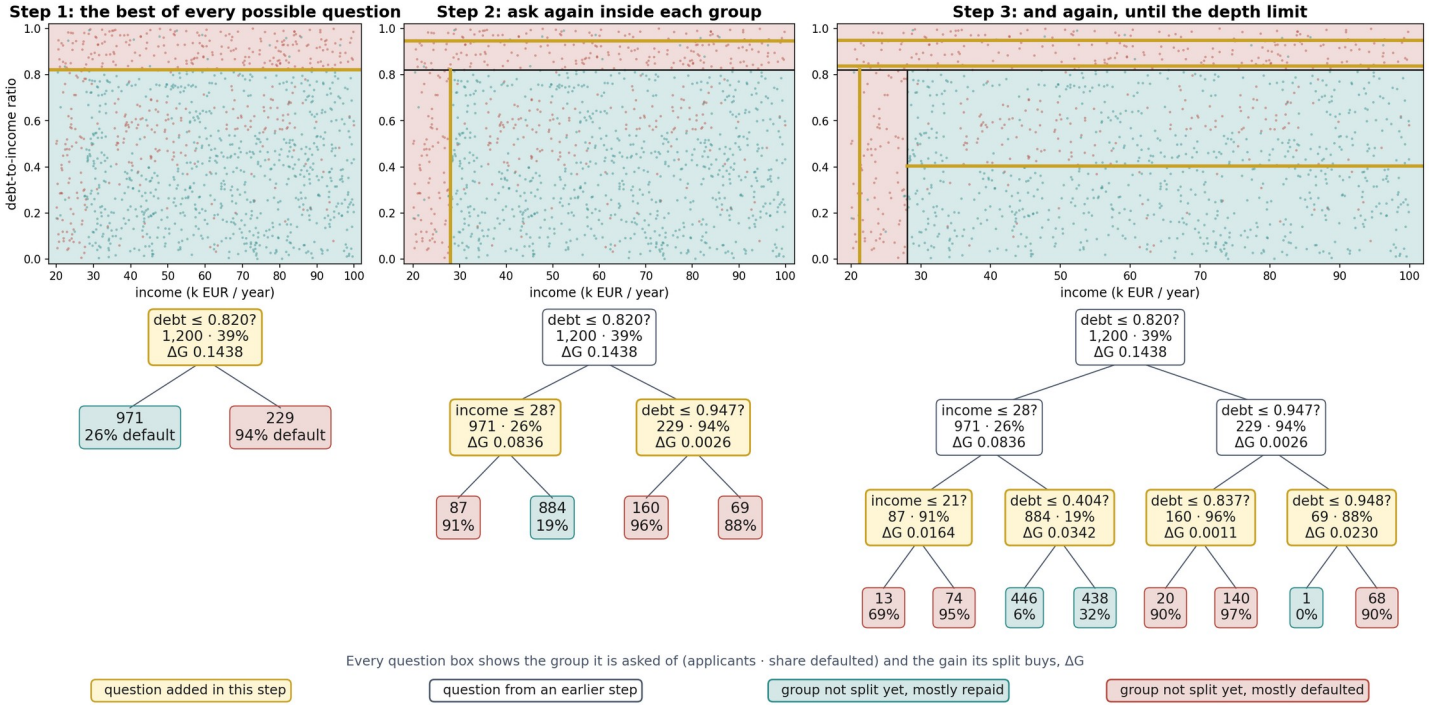
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The whole algorithm, in one sentence

- Try **every feature and every threshold**; keep the split leaving the **purest** two groups
- Apply it; repeat on each group, independently
- Stop at a depth limit, a pure group, or too few examples to split
- No coefficients, no gradient: thresholds are found by **searching**
- **Greedy**: whichever split helps most *right now*, with no lookahead

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The algorithm, drawn



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How pure is a group of examples?

- A leaf that is all one class needs no more questions
- A group half one class, half the other, is the worst case: a coin flip
- **Impurity** is a number that captures this: 0 for a pure group, highest for an even mix. “As pure as possible” then means “with the lowest impurity”

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Gini impurity

$$G = 1 - \sum_{c=1}^{C} p_c^2$$

p_c : share of class c (defaulted, repaid)

slide 8's 229 : $1 - 0.939^2 - 0.061^2 = 0.115$

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For two classes, one number

- G = 2p(1-p), where p is the share defaulted and 1-p the share repaid
- **0** when the group is pure (p = 0 or p = 1)
- **0.5**, the maximum, at p = 0.5
- On the full 1,200 applicants, p = 0.387 gives G = 0.474

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The gain a split buys

- A split divides a parent P into a left child L and a right child R; *m* counts each one's examples: at the root, 1,200 in P = 971 in L + 229 in R
- Worth taking if the children's impurities, **averaged**, are lower than the parent's: at the root 0.474 against 0.330
- The average is **weighted by size**: one example cut off alone would halve a plain average, and barely moves a weighted one

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The reduction in impurity

$$\Delta G = G_P - \left(\frac{m_L}{m_P} G_L + \frac{m_R}{m_P} G_R \right)$$

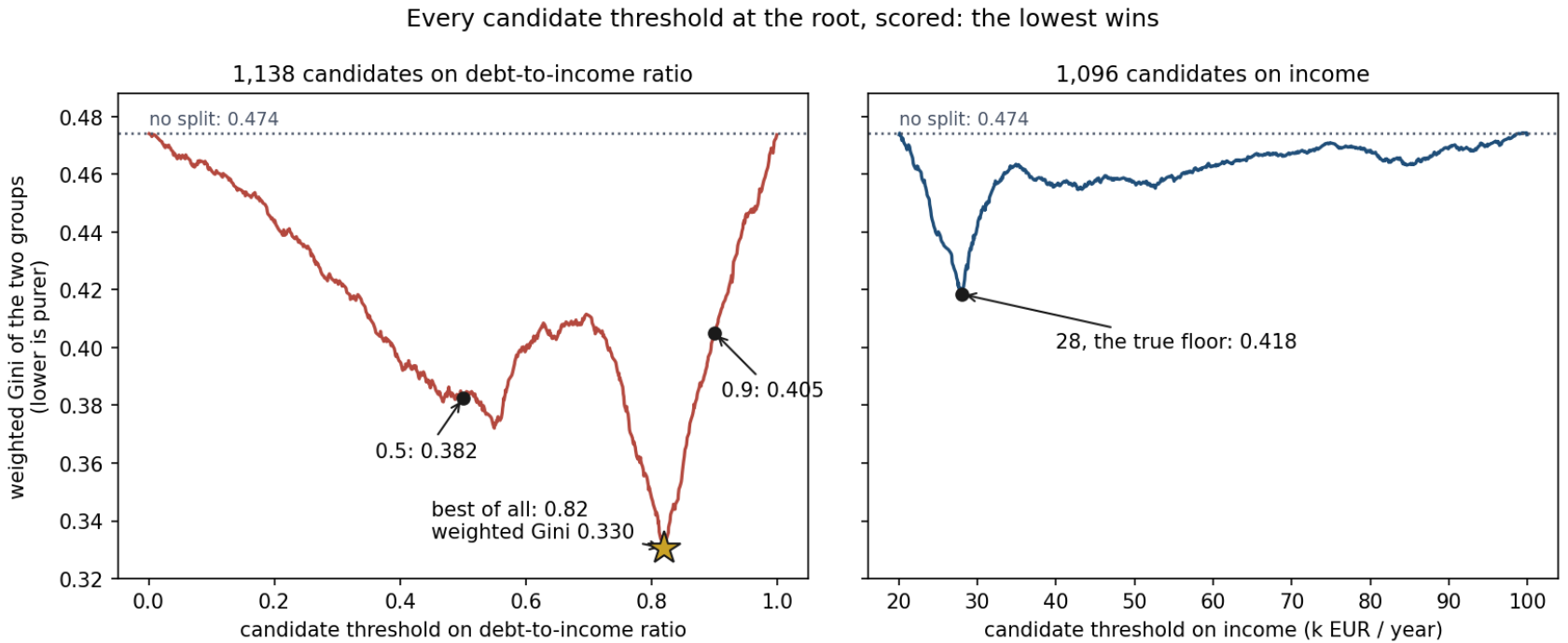
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Scanning for the best split

- Sort a feature’s values: every **midpoint** between neighbours is a candidate threshold
- Infinitely many thresholds, but **finitely many splits**: any threshold between the same two neighbours makes the same two groups
- Score every candidate on **every feature**; keep the best pair, then repeat inside each child
- **Greedy**: ΔG is judged one split at a time, never jointly

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Every threshold, scored



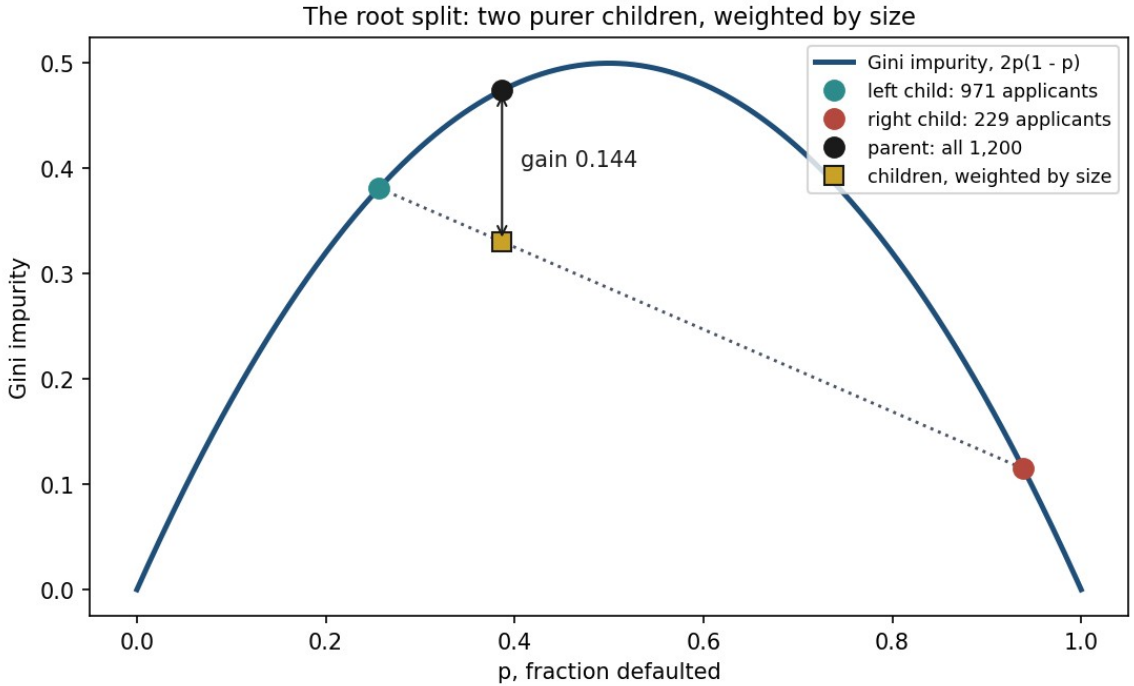
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Worked example: the root split

- Parent: 464 of 1,200 defaulted, G = **0.474**; the search finds `debt_ratio <= 0.82`, the **debt ceiling**
- Left: 249 of 971 defaulted, p = 0.256, G = 2 × 0.256 × 0.744 = **0.381**
- Right: 215 of 229, p = 0.939, G = 2 × 0.939 × 0.061 = **0.115**
- Weighted: 0.809 × 0.381 + 0.191 × 0.115 = 0.330; gain 0.474 - 0.330 = **0.144**

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The root split, on the Gini curve



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The loop, in seven lines

```
def grow(g):
  if stop(g): return leaf(g)
  gain = {c: ΔG(g, c) for c in splits(g)}
  best = max(gain, key=gain.get)
  if gain[best] <= 0: return leaf(g)
  L, R = split(g, best)
  return node(best, grow(L), grow(R))
```


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Checked against scikit-learn

- The from-scratch implementation and scikit-learn's `DecisionTreeClassifier` make **the same prediction for all 1,200 applicants**, at every depth tested
- Not a coincidence: both search the same space by the same greedy rule
- From here on, the notebooks use scikit-learn's implementation: the same algorithm, compiled, not a different one

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What greedy misses

- The search never asks “which pair of splits helps most together”: only “which single split helps most right now”
- A feature that only pays off **in combination** with a later split can be invisible to a shallow tree
- Exactly what happens to this dataset’s two interior “stressed” islands, covered next

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Depth is the bias-variance dial

- `max_depth`: the most questions **one applicant** answers
- **Shallow tree**: few questions, only the coarsest structure. High bias, low variance
- **Deep tree**: enough questions to isolate single points. Training accuracy **exactly 1.000**, even on data with no structure, unless two examples share every feature value but not the label

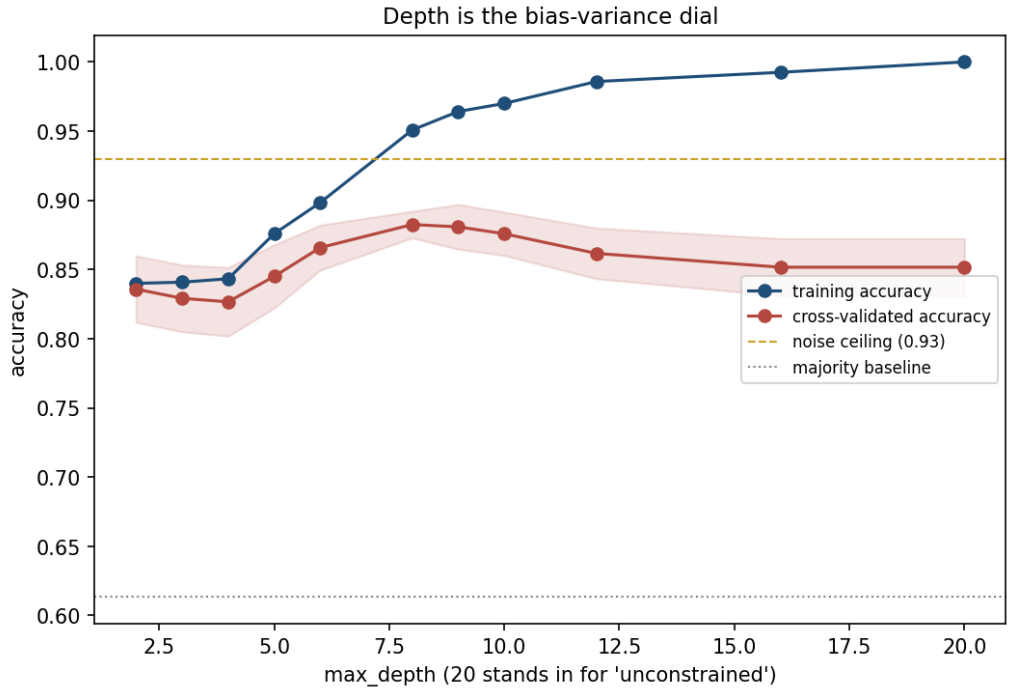
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Measured, at seven depths

max_depth	Training accuracy, all 1,200 loans	Cross-validated, 5 folds of the 1,200 loans (mean ± standard deviation)
2	0.840	0.836 ± 0.024
4	0.843	0.827 ± 0.025
6	0.898	0.866 ± 0.016
8	0.951	**0.882 ± 0.010**
12	0.986	0.862 ± 0.018
16	0.993	0.852 ± 0.021
unconstrained	**1.000**	0.852 ± 0.021

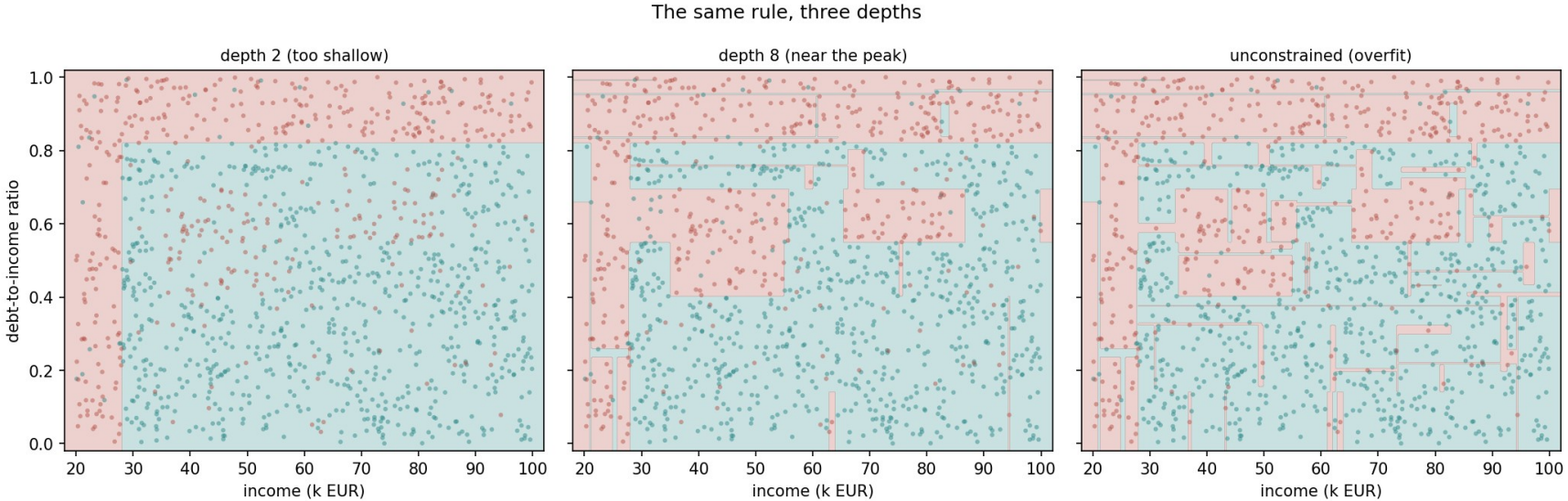
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The curve bends downward



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The same rule, three depths



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Reading a tree, and the knob that keeps it small

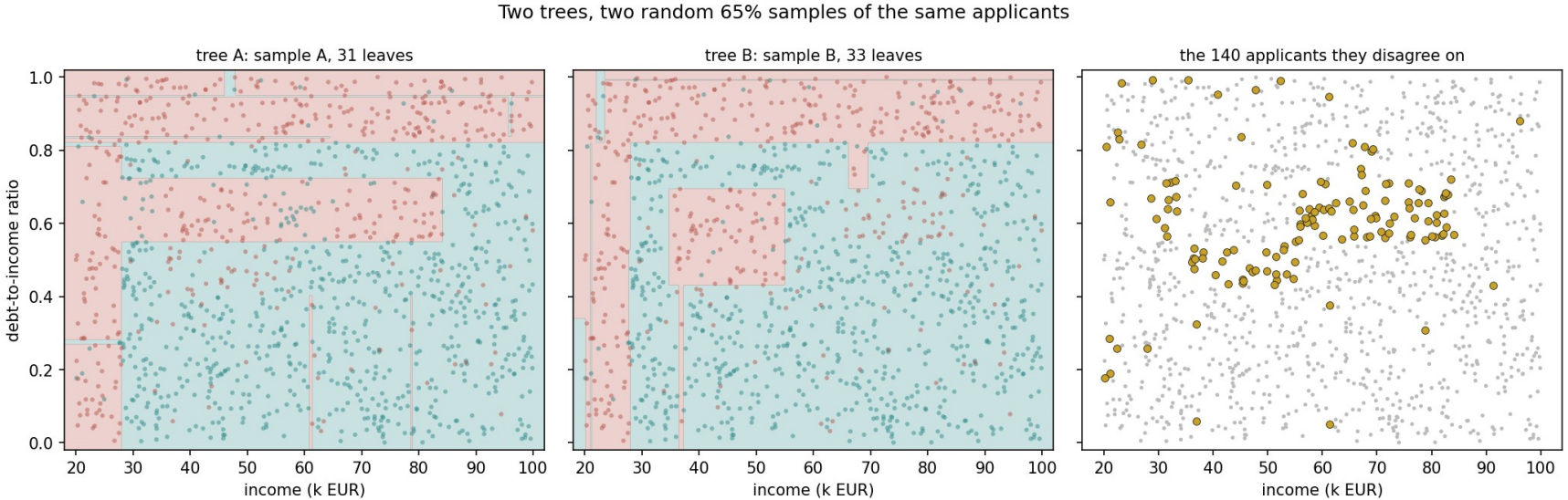
- `max_depth` **caps every path**: each node above the cap is split if it can be, worth it or not
- `max_leaf_nodes` is a **budget of leaves**, spent first where ΔG × share of applicants is largest
- `max_leaf_nodes = 9`: **0.908** training, **0.901** cross-validated, above depth 8's 0.882; thresholds 28 and 0.82
- In code: `export_text(tree)` prints the tree as if/else rules

A tree that doesn’t stay still

- Two trees at `max_depth = 6`, each on 780 of the 1,200 applicants, drawn **without replacement** (no one twice)
- The root, decided on all 780, agrees to within **0.001**: the debt ceiling
- The 29 or so questions below it, each on fewer applicants, do not: **31 and 33 leaves**, **opposite verdicts for 140** of the 1,200 (11.7%)

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Two trees, 140 different verdicts



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Notebook 1, live

section, and its slides	you call (→ the number to come back with)
1. look at it first, 4-6	`make_pipeline(StandardScaler(), LogisticRegression()), cross_val_score, StratifiedKFold` → 0.613 against 0.748
2. a tree, split by split, 7-20	your own `gini` and `best_split`, then `DecisionTreeClassifier(max_depth=d)` → 100.0% identical predictions; gain 0.1438
3. the depth dial, 21-24	`DecisionTreeClassifier(max_depth=d)` in `cross_val_score` → peak 0.882 at depth 8
4. what a tree costs, 25-27	`DecisionTreeClassifier(max_leaf_nodes=9), export_text` → 0.908 training, 0.901 cross-validated; two trees differ on 140
or atoms 1-7	`Consolidation/07_lesson_7_atoms.ipynb`

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Break

- Twelve minutes

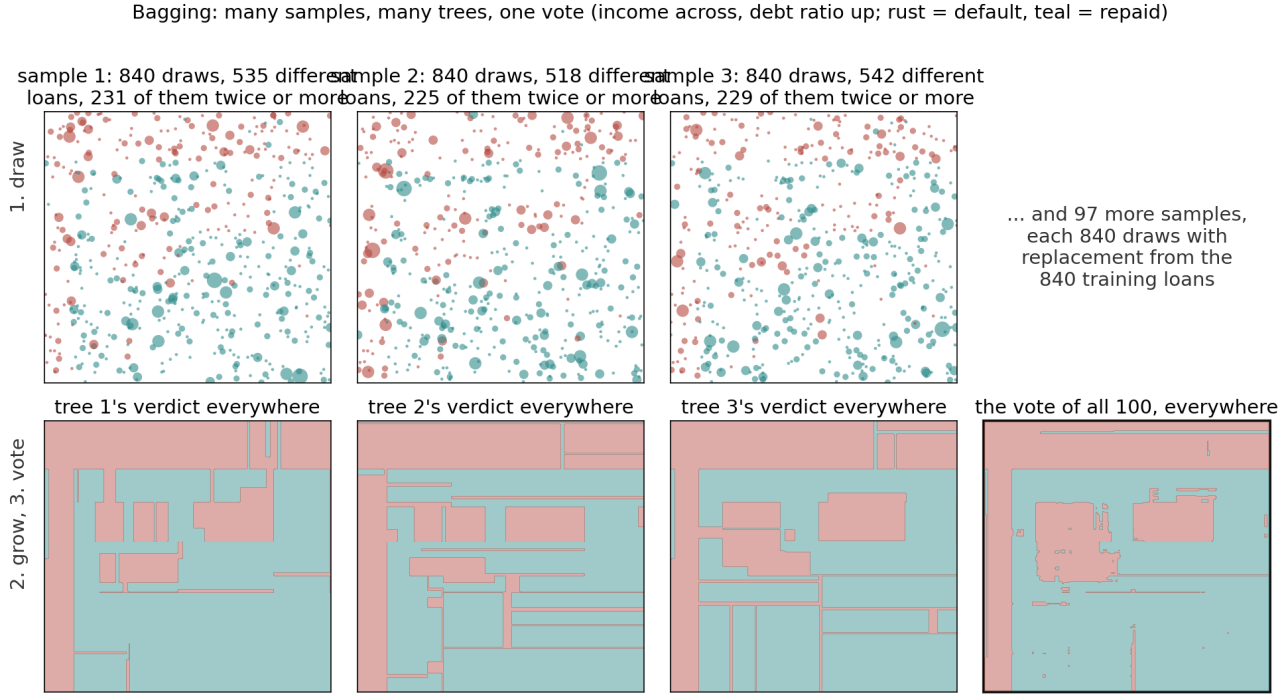
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Bagging: many trees, one vote

- Slide 27: two trees on samples of the same applicants disagreed on 140 of 1,200
- So train **many trees**, each on its own sample, and **vote**: one sample's bad luck is outvoted
- Each sample: m rows drawn **with replacement** from the m training rows, some twice, some never: a **bootstrap sample**
- A tree on each sample, then a majority vote: **bagging**

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Bagging, drawn



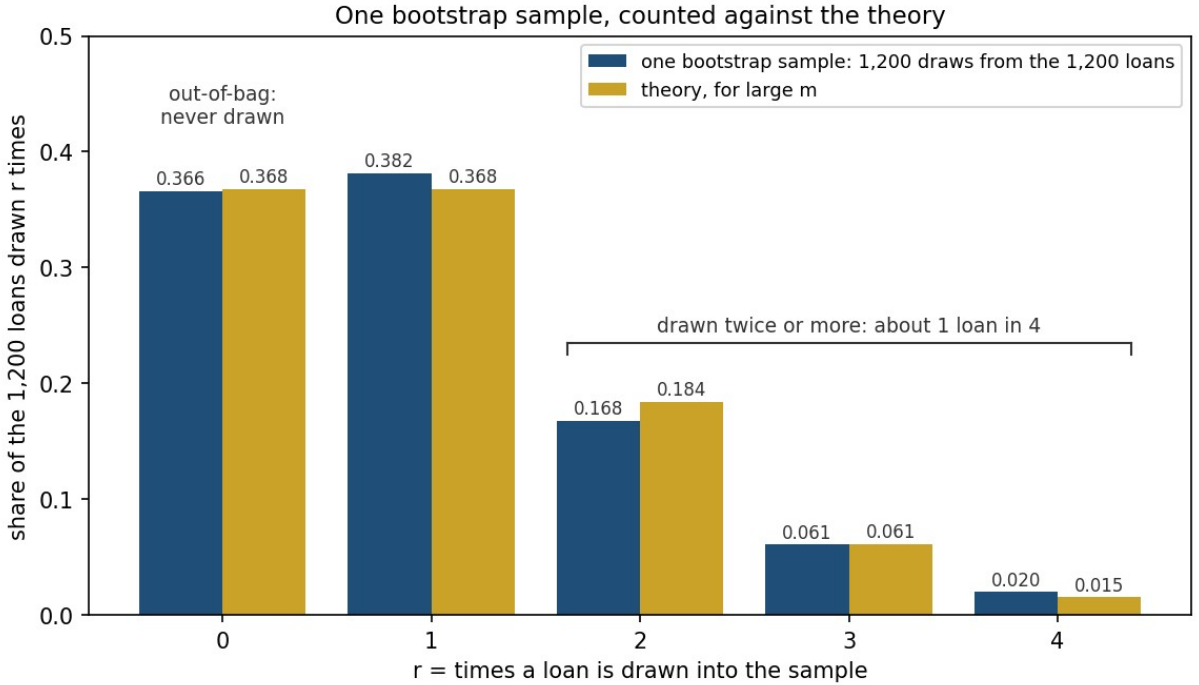
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How much data one bootstrap sample sees

- A draw picks a given row with probability 1/m, so it misses it with 1 - 1/m; all m independent draws miss it with $(1 - 1/m)^m$, about $e^{-1} = 0.368$
- So each sample holds about **63.2%** of the different rows; the other **36.8%** are that tree's **out-of-bag (OOB)** rows, which it never sees
- Still m draws: at m = 1,200, on average 1,200 × (1 - 0.3677) = 759 different rows, and 441 repeated draws

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One bootstrap sample, counted against the theory



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Out-of-bag: each loan judged by the trees that never saw it

Out-of-bag: each loan is judged only by the trees that never saw it

	tree 1	tree 2	tree 3	tree 4	tree 5	tree 6	tree 7	tree 8	tree 9	tree 10	out-of-bag verdict	true label	
loan 0	seen ×1	seen ×2	seen ×4	seen ×1	seen ×1	seen ×1	seen ×1	seen ×1	**D**	seen ×1	**D**	**D**	right
loan 1	seen ×1	seen ×1	seen ×3	**R**	seen ×2	**D**	seen ×1	**D**	**D**	seen ×3	**D**	**R**	wrong
loan 2	seen ×1	**D**	seen ×3	seen ×1	seen ×2	seen ×2	seen ×1	seen ×1	**D**	seen ×1	**D**	**D**	right
loan 3	seen ×1	seen ×4	**R**	seen ×2	**R**	**R**	**R**	seen ×2	seen ×2	**R**	**R**	**D**	wrong
loan 4	seen ×1	seen ×1	seen ×1	seen ×2	seen ×3	**D**	seen ×3	seen ×1	seen ×1	seen ×2	**D**	**D**	right
loan 5	seen ×2	**R**	seen ×1	seen ×2	seen ×2	seen ×1	**R**	**R**	seen ×2	seen ×2	**R**	**R**	right
loan 6	seen ×4	seen ×2	seen ×1	seen ×4	seen ×1	**D**	seen ×1	**D**	**D**	seen ×3	**D**	**D**	right
loan 7	**R**	seen ×1	**R**	seen ×1	seen ×1	seen ×2	**R**	seen ×1	**R**	seen ×1	**R**	**R**	right

grey: the loan is in that tree's sample (×2 = twice), so that tree does not vote on it; gold: the tree never saw it and votes - D defaulted, R repaid

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Out-of-bag against cross-validation

- How accurate is a bagging of 300 trees on loans **it never saw**? All 1,200 loans train it
- **Out-of-bag**: each loan judged by the trees that never saw it (about 0.368 × 300 = 110): **1,090** right, accuracy **0.9083**
- **5-fold cross-validation** (lesson 5, next slide): **1,083** right, accuracy **0.9025** ± 0.021
- They agree within 0.006; out-of-bag needs **no loans set aside, no extra training**

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Cross-validation, step by step: five groups of 240

Round: the group left out	Right of its 240	Accuracy = right ÷ 240	Deviation from the mean of the five = accuracy − 0.9025	Deviation²
1	212	0.8833	−0.0192	0.000367
2	215	0.8958	−0.0067	0.000044
3	223	0.9292	+0.0267	0.000711
4	222	0.9250	+0.0225	0.000506
5	211	0.8792	−0.0233	0.000544
Mean = sum ÷ 5	1,083	**0.9025**		sum 0.002174
Standard deviation = √(0.002174 ÷ 5)		**0.021**		

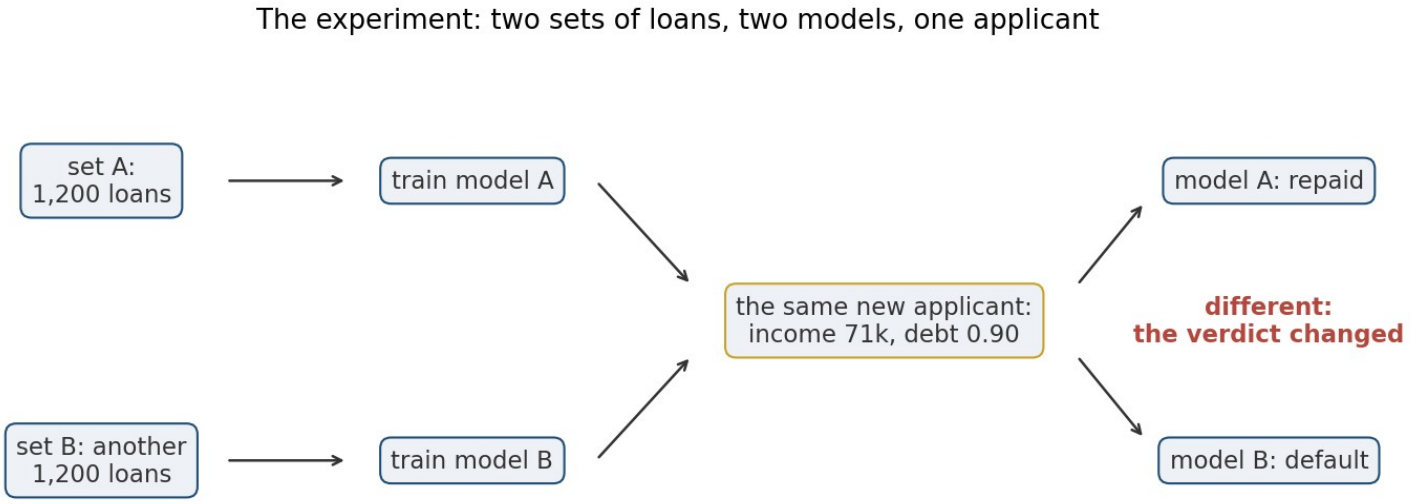
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Why voting helps, and its floor

- A tree’s verdict has **two parts**: what the 1,200 loans say, shared by all trees, and the **luck of its own sample**
- The luck differs from tree to tree: in a vote it **cancels**
- If the 1,200 loans mislead, every tree is misled **alike**: the vote **cannot cancel it**
- So more trees help only down to a **floor**; lower needs trees **less alike**

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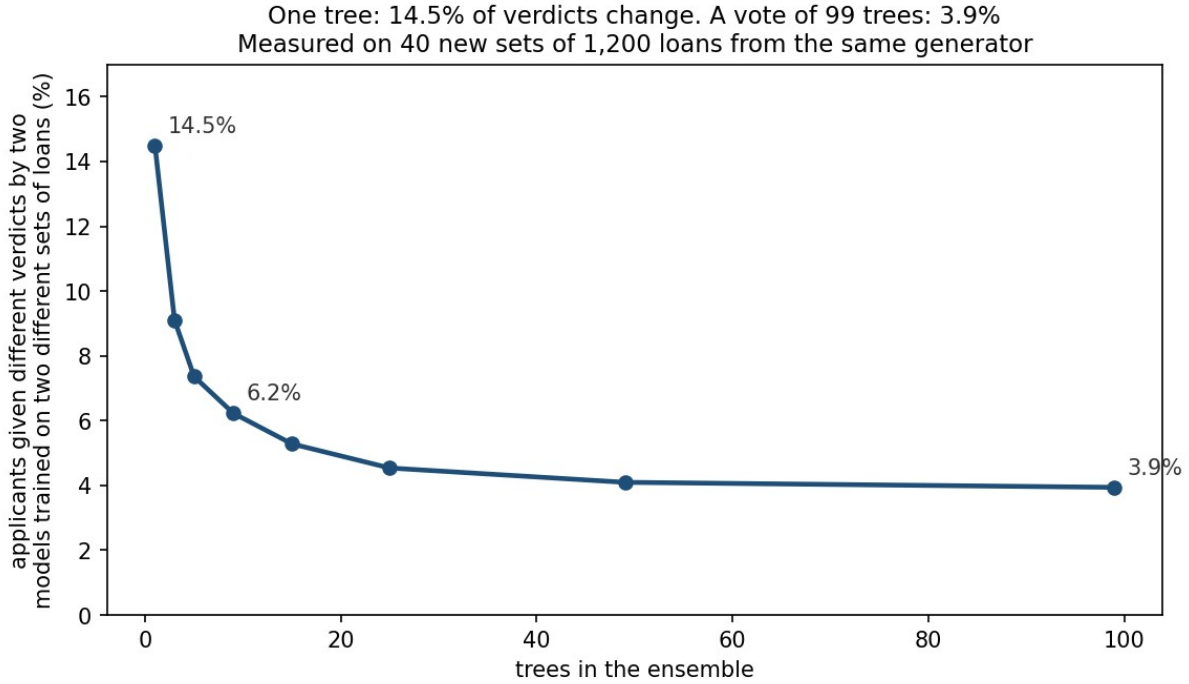
The experiment: two sets of loans, two models, one applicant



Repeat for 300 applicants and for every pair of the 40 sets of loans: on what share of applicants do the two models disagree?

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Another 1,200 loans: how many verdicts change?

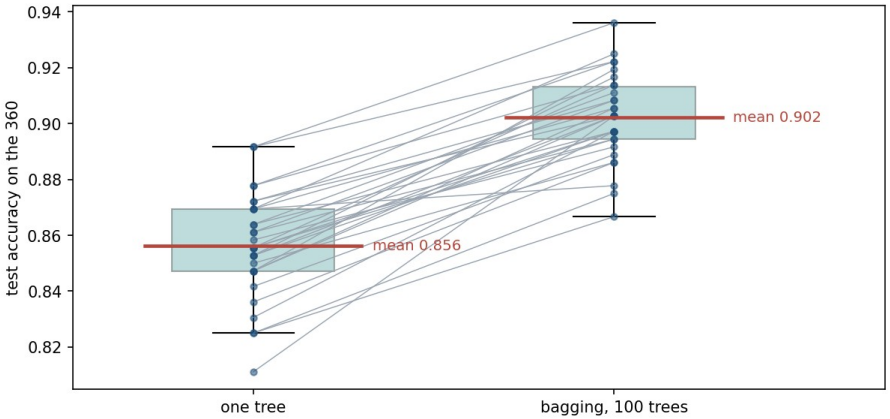


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Bagging against one tree, on 30 splits



repeated with 30 different random splits; a grey line joins the two models on the same split



box: the middle half of the 30 splits; red line: the mean; dots: the splits

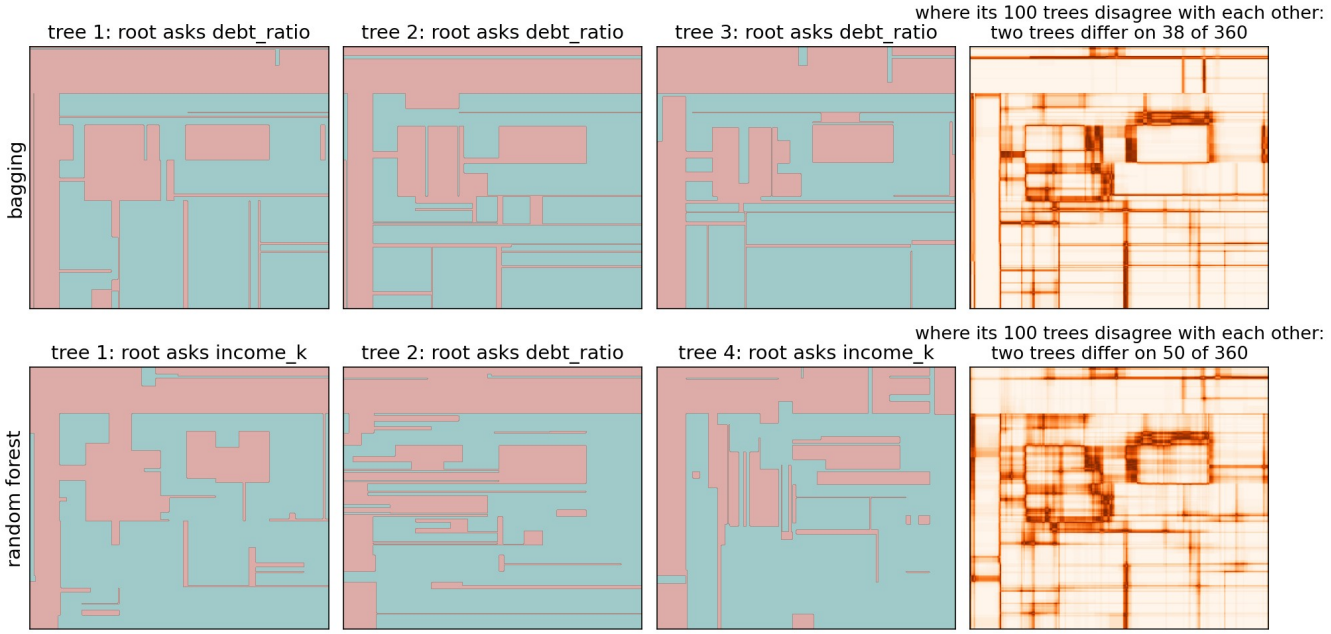
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Random forests: trees less alike

- Bagging: every split may use **every column**, so all 100 trees start on `debt_ratio`
- **Random forest**: bagging, but each split may use only `max_features` random columns, **redrawn at every split**
- Default √n rounded down, n = number of **columns**; here n = 2: **one** per split
- A tree is **worse** alone (0.828 vs 0.837), the forest **better**

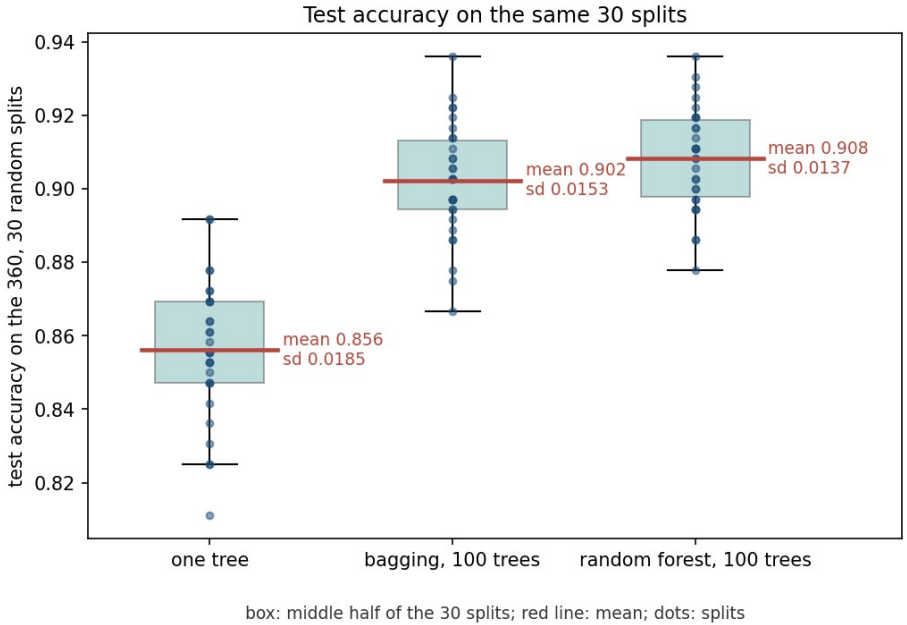
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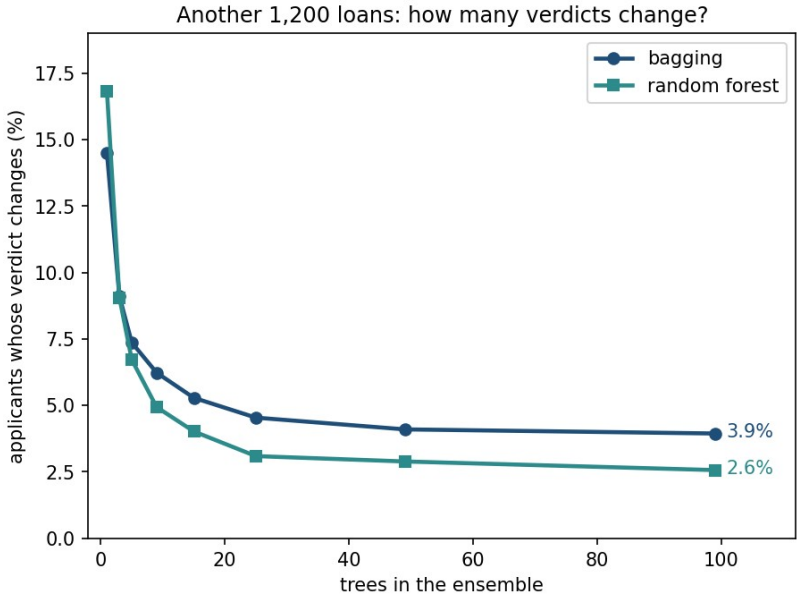
Bagged trees against forest trees



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Random forests, measured





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Feature importance on one tree

Split of slide 8’s tree	Applicants	Gain ΔG	Weight = applicants ÷ 1,200	Credit = ΔG × weight
debt ratio ≤ 0.82 (root)	1,200	0.1438	1.0000	0.1438
income ≤ 28 (left)	971	0.0836	0.8092	0.0676
debt ratio ≤ 0.947 (right)	229	0.0026	0.1908	0.0005
Debt: 0.1438 + 0.0005				0.1443 ÷ 0.2119 = **0.681**
Income				0.0676 ÷ 0.2119 = **0.319**

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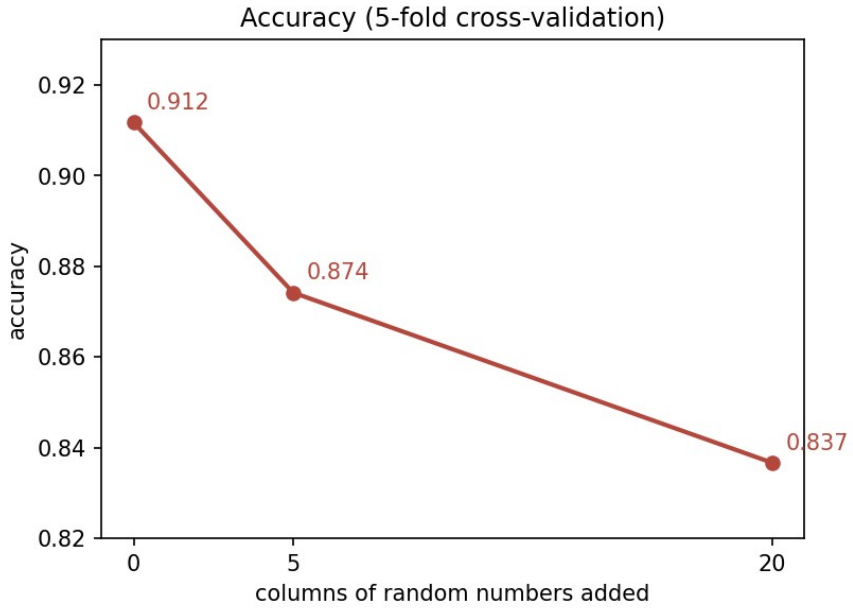
What a forest’s importance says

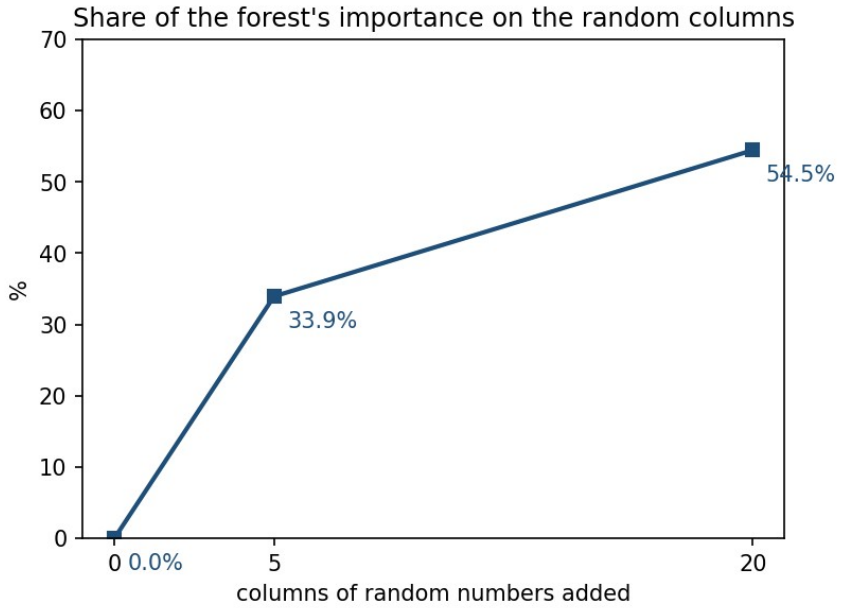
- **Feature importance** says how much each column helped the trees’ questions, scaled so that all the columns add up to 1
- A 300-tree forest on the loans: **debt 0.61, income 0.39**
- **Predict**: add columns of random numbers, unrelated to default. How much importance should they get?

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More columns, more noise

A 300-tree random forest on the loans, with 0, 5 or 20 columns of random numbers added





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Permutation importance: the procedure

Permutation importance: shuffle one column, see how much worse the model gets

applicant	income	debt ratio		debt ratio, shuffled
1	32.9	0.80		0.98
2	39.2	0.97		0.98
3	24.6	0.20	shuffle →	0.05
4	72.8	0.46		0.09
5	61.6	0.63		0.44

accuracy before: 0.806

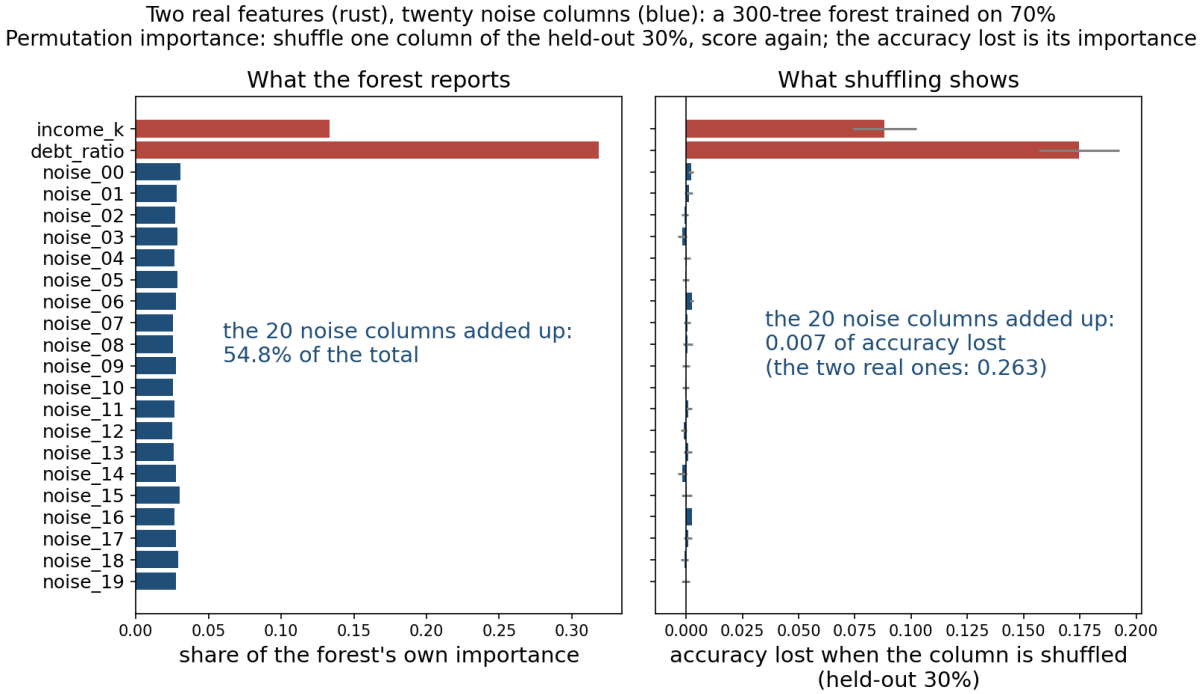
accuracy after: 0.600

drop 0.206
= importance of debt ratio

1. trained forest, 30% held-out applicants 2. shuffle one column among them 3. score again 4. repeat for every column

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Permutation importance: claimed against measured



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Notebook 2, live

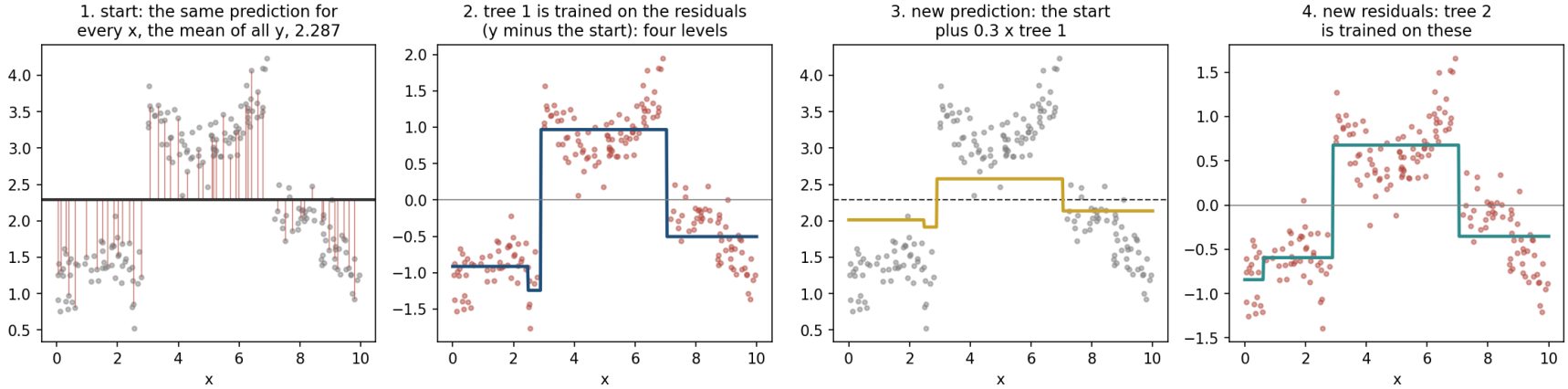
section, and its slides	you call (→ the number to come back with)
1. bagging from scratch, 30-31	`train_test_split`, your bagging and `BaggingClassifier`, `accuracy_score` → 0.872 both; 1 of 29 flipped right
2. why averaging works, 32-39	`oob_score=True`: `.oob_score_` against `cross_val_score` → 0.9083, 0.9025
3. variance, measured, 40, 43	tree, bagging, `RandomForestClassifier`, 30 splits → 0.856, 0.902, 0.908
4. what a forest adds, 41-42	`tree_.feature[0]` → roots 100, 51, 49
5. feature importance, 44-46	`.feature_importances_`, 0 to 20 noise columns → 54.5% on noise
6. checking the importances, 47-48	`permutation_importance`, test split → real 0.09 and 0.18, noise ≈ 0
or atoms 8-13	`Consolidation/07_lesson_7_atoms.ipynb`

Boosting: one tree after another

- Bagging and random forests grow trees **independently**, then let them vote
- **Gradient boosting** grows trees **in sequence**: each is trained to correct what the trees before it still get wrong
- Each tree is shallow and outputs a **number**, a correction, not a class: a **regression tree**
- The learning rate α decides **how much** of each correction is added

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Boosting, drawn: the first two steps



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Building the ensemble, one tree at a time, for squared error

$F_0(x) = \bar{y}$: start from the mean

$F_t(x) = F_{t-1}(x) + \alpha\, h_t(x)$: add a scaled tree

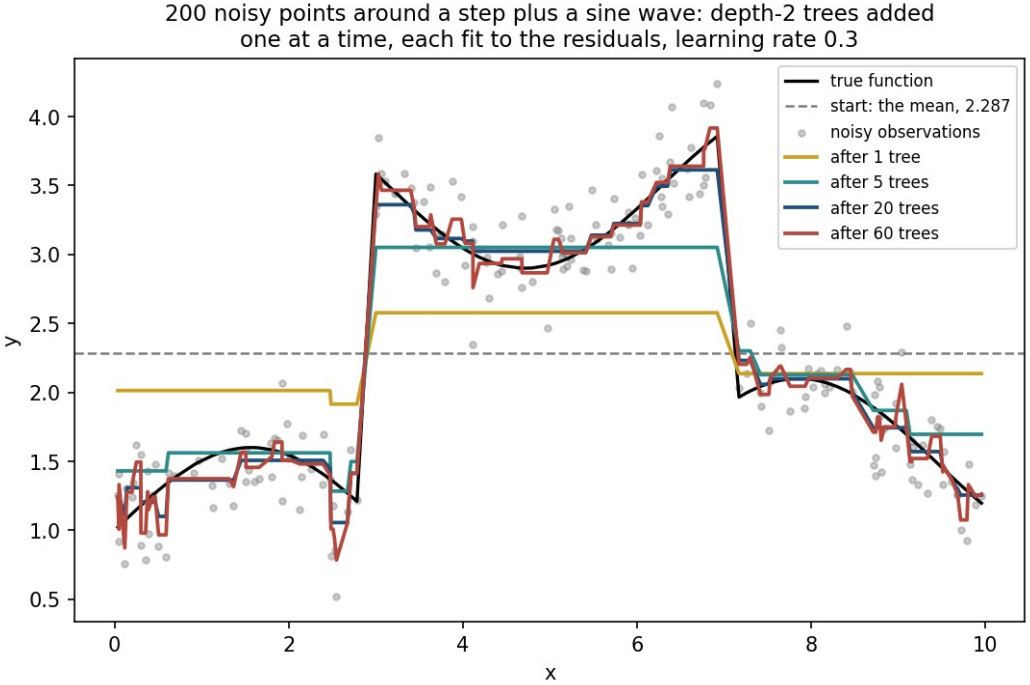
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What each tree is fit to

- For squared error, the next tree h_t is trained on the **residual** $y - F_{t-1}(x)$: what the trees so far have not explained
- In general, h_t is trained on the **negative gradient** of the loss with respect to the prediction: where each prediction should move
- With $L = \frac{1}{2}(y - F)^2$, $-\partial L/\partial F = y - F$: the same thing as the residual

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A step plus a sine wave, fit by 60 shallow trees



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Training error keeps falling; the true error does not

Trees	Mean squared error (MSE) on the 200 training points	MSE against the true function
1	0.476	0.395
5	0.119	0.068
20	0.046	**0.012**, the best point
60	**0.025**, still falling	0.018, rising: it tracks noise

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Classification: the same idea, a different gradient

- Now y is 0 or 1. The running score F becomes a probability through lesson 4's logistic function, $p = \sigma(F)$
- The loss is lesson 4's **log-loss**; its negative gradient is y - p, the **pseudo-residual**, lesson 4's error term: each tree is trained on it
- Start: $p = 464/1{,}200 = 0.387$ for all (the share who defaulted), so the first tree fits $+0.613$ to defaulters and -0.387 to the rest

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On the loan data

- `GradientBoostingClassifier`, defaults: **100 trees** of depth 3, learning rate $\alpha = 0.1$
- **0.902 ± 0.020** cross-validated accuracy: all 1,200 loans, the same five groups of 240 as slide 36
- Ahead of the depth-8 tree (0.882), level with the nine-leaf tree (0.901)

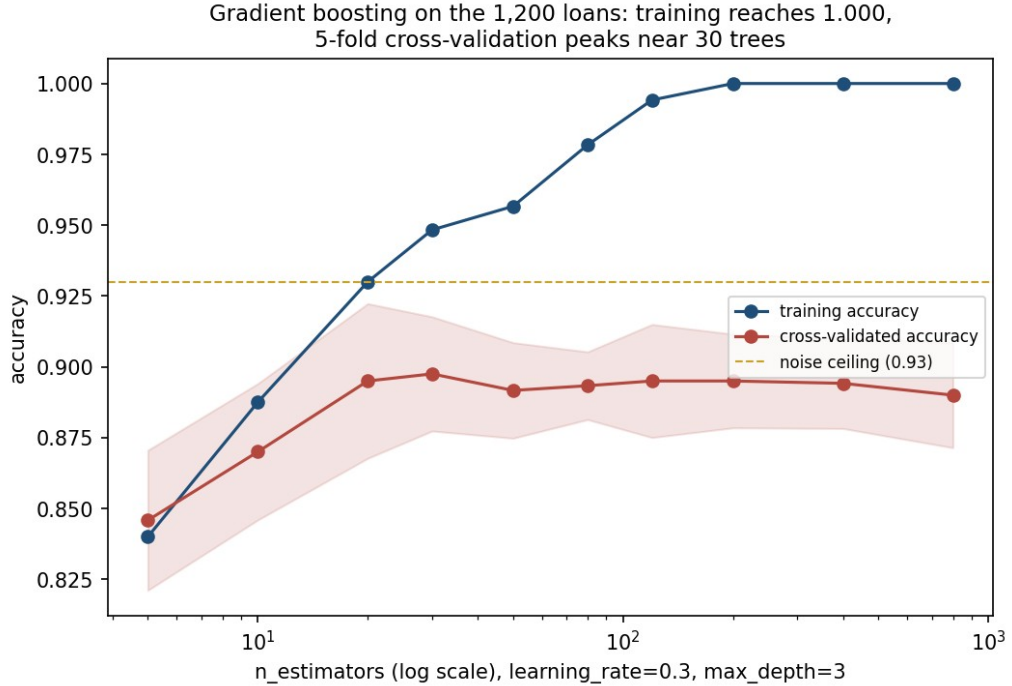
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Boosting never stops fitting on its own

`n_estimators` (α = 0.3, depth 3)	Training accuracy, all 1,200 loans	Cross-validated, 5 folds of the 1,200 loans
5	0.840	0.846 ± 0.025
10	0.888	0.870 ± 0.024
20	0.930	0.895 ± 0.027
30	0.948	**0.897 ± 0.020**
80	0.978	0.893 ± 0.012
200	1.000	0.895 ± 0.017
800	1.000	0.890 ± 0.019

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Training climbs to 1.000 and stays; the honest curve turns over first



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Fixing 120 trees, varying only the learning rate

Learning rate α	Training accuracy, all 1,200 loans	Cross-validated, 5 folds of the 1,200 loans
0.02	0.888	0.872 ± 0.025
0.05	0.914	0.897 ± 0.022
0.10	0.939	**0.902 ± 0.022**
0.30	0.994	0.895 ± 0.020
1.00	1.000	0.878 ± 0.020

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Notebook 3, live

section, and its slides	you call (→ the number to come back with)
1. boosting from scratch, 50-55	`DecisionTreeRegressor(max_depth=2)` on the residuals, 60 times → MSE 0.0119 at 20 trees
2. classification, 56-57	`GradientBoostingClassifier()` in `cross_val_score` → F0 = −0.461; 0.902
3. more trees, 58-59	the same, `learning_rate=0.3`, `n_estimators` from 5 to 800 → peak 0.897 at 30
4. learning rate, 60	`n_estimators=120`, `learning_rate` from 0.02 to 1.0 → best 0.902 at 0.1
5. leaderboard, 62	every model in one `cross_val_score` loop → forest 0.911
or atom 14	`Consolidation/07_lesson_7_atoms.ipynb`

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Every method, on the same five folds

Model, on the 1,200 loans	Cross-validated accuracy	± over the 5 folds
Majority baseline	0.613	
Logistic regression	0.748	0.030
Single tree, unconstrained	0.852	0.021
Single tree, `max_depth = 8` (chosen on these folds)	0.882	0.010
Gradient boosting, 30 trees (chosen on these folds)	0.897	0.020
Single tree, nine leaves	0.901	0.020
Bagging, 100 trees	0.904	0.020
Random forest, 100 trees	**0.911**	0.018
noise ceiling	*≈ 0.93*	

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Three findings

- **On this dataset** the three ensembles land within 1.5 points of each other, and a **nine-leaf tree** among them (0.901): one a person can read
- What the ensembles buy here is **steadiness**: over 30 splits, the nine-leaf tree’s spread is 0.020, the forest’s 0.0137
- The **unconstrained tree is the worst tree**, though the most flexible

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How to choose, in practice

Situation	Reach for
The decision must be explained to the person it affects	A single tree, small enough to read; if even that is too opaque, none of these models
Tabular data, accuracy matters most, no explanation required	A random forest: the best score here, at its default settings
Accuracy matters, and there is time to tune the number of trees and the learning rate together	Gradient boosting, tuned on validation data
Many columns, few of them informative	Any tree method, but check its importances by permutation
Too few rows to set a validation set aside	A random forest, judged by its out-of-bag score

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54% of the importance on noise

- 20 random columns, 2 real ones: over half the importance lands on the random ones
- **The menu**: each split may use 4 of the 22 columns; C(20,4) / C(22,4) = 4,845 / 7,315 = 0.662: two splits in three see no real one
- **Small groups**: with all 22 on offer, noise still takes **28%**: on a few applicants deep in a tree, a random column fits a little by chance
- The check: **permutation importance**

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What to take away

- A tree asks **one question at a time**; **depth** is the dial; one tree alone is **unstable**
- Trees that **vote** cancel bad luck; a forest makes them **less alike**
- **54%** of a forest’s importance on noise: check by **shuffling**
- **Boosting** chains trees, each fixing the rest; **stop it** on unseen data

Homework

- **Exercise 7**, discussed at the start of **Friday 4 December**
- `Exercises/07_trees_and_ensembles.md`

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